Inspection Report Analysis

Food Establishment Inspection Forms

Chicago & Dallas

Source-to-Dataset Field Mapping & Violation Code Reference

DAMG7370: Designing Advanced Data Architectures for Business Intelligence
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1. Purpose

This document analyzes the official food establishment inspection report forms used by Chicago (Illinois) and Dallas (Texas), maps their fields to the corresponding columns in the source datasets, and documents how each field is captured, transformed, and loaded into the dimensional model. Understanding the inspection reports is critical for interpreting the data correctly and designing appropriate cleansing and transformation logic.

2. Chicago Inspection Report

The Chicago Food Establishment Inspection Report is issued by the Chicago Department of Public Health, Food Protection Division. The form captures establishment details, inspection metadata, and a comprehensive 47-item checklist of violation codes organized by severity.

2.1 Report Structure

The Chicago inspection report consists of two pages. Page 1 contains the establishment header (identification, licensing, and facility details), food temperature observations, and the 47-item violation checklist. Page 2 captures certified manager details, a free-text area for specific code violations observed and correction deadlines, HACCP principles, inspector comments, and the final inspection result (Passed, Passed w/Cond, Failed, or Canceled).

2.2 Issuing Authority

Chicago Department of Public Health, Food Protection Division. The form references Illinois state law requiring inspection under a certified environmental health sanitarian. Contact: 312.746.8030.

2.3 Report Fields to Dataset Mapping

The following table maps every field on the Chicago inspection report form to its corresponding column in the Chicago Open Data Portal dataset.

Report Field	Dataset Column	Data Type	Notes
Inspection #	Inspection_ID	integer	Unique identifier for each inspection
License #	License	integer	Business license number
Inspection Date	Inspection_Date	date	Date the inspection was conducted
Inspection Type	Inspection_Type	string	Canvass, License, Complaint, Re-Inspection, etc.
Facility Type	Facility_Type	string	Restaurant, Grocery Store, School, Bakery, etc.

D/B/A (Doing Business As)	DBA_Name	string	Primary business name
A/K/A (Also Known As)	AKA_Name	string	Alternate business name
Business Address	Address	string	Street address of establishment
Zip	Zip	integer	Zip code of establishment
City	City	string	City name (always CHICAGO)
State	State	string	State abbreviation (always IL)
Risk	Risk	string	Risk 1 (High), Risk 2 (Medium), Risk 3 (Low)
Code Violations	Violations	string	Pipe-delimited string with all violation codes, descriptions, and comments
PASSED / PASSED w/COND / FAILED / CANCELED	Results	string	Final inspection outcome
Latitude (geocoded)	Latitude	double	Not on form; geocoded from address
Longitude (geocoded)	Longitude	double	Not on form; geocoded from address
Location (geocoded)	Location	string	Combined lat/long; not on form
Inspector Badge #	Not captured	—	On form but not in dataset
Supervisor Badge #	Not captured	—	On form but not in dataset
Fire #	Not captured	—	On form but not in dataset
SR # / SFP #	Not captured	—	On form but not in dataset
Certified Manager	Not captured	—	On form but not in dataset
Certificate #/Type	Not captured	—	On form but not in dataset
# Employees / # Seats	Not captured	—	On form but not in dataset
Food Temperature Observations	Not captured	—	On form but not in dataset
HACCP Principle	Not captured	—	On form but not in dataset

2.4 Chicago Violation Code Reference

The Chicago inspection form uses a 47-item numbered checklist organized into three severity tiers. Violations are recorded in the dataset as a pipe-delimited string in the Violations column. Each violation follows the format: "CODE. DESCRIPTION - Comments: COMMENT_TEXT".

Critical Violations (Items 1–14) — $500 fine per violation

Code	Category	Description
01	Food Protection 7-38-005 (B)(B-2)	Source sound condition, no spoilage, foods properly labeled, shellfish tags in place
02	Food Protection 7-38-005 (A)	Facilities to maintain proper temperature
03	Food Protection	Potentially hazardous food meets temperature requirement during storage, display, and service
04	Cross Contamination	Source of cross contamination controlled (cutting boards, food handlers, utensils)
05	Food Protection 7-38-010 (A)(B)	Personnel with infections restricted; no open sores, wounds
06	Hand Hygiene	Hands washed and cleaned, good hygienic practices; bare hand contact minimized
07	Equipment Sanitation 7-38-030	Wash and rinse water: clean and proper temperature
08	Equipment Sanitation	Sanitizing rinse for equipment and utensils: clean, proper temperature, concentration, exposure time
09	Waste Water Disposal 7-38-030	Water source: safe, hot & cold under city pressure
10	Waste Water Disposal	Sewage and waste water disposal, no back siphonage, cross connection and/or back flow
11	Toilet/Hand Washing 7-38-030	Adequate number, convenient, accessible, designed, and maintained
12	Hand Washing Facilities	With soap and sanitary hand drying devices, convenient and accessible to food prep area
13	Insect/Rodent Control 7-38-020	No evidence of rodent or insect infestation, no birds, turtles, or other animals
14	Compliance	Previous serious violation corrected (7-42-090)

Serious Violations (Items 15–29) — $250 fine per day until corrected

Code	Category	Description
15	Food Protection 7-38-005 (A)	Unwrapped and potentially hazardous food not re-served

16	Food Protection	Food protected during storage, preparation, display, service and transportation
17	Food Safety	Potentially hazardous food properly thawed
18	Insect/Rodent Control: Garbage	No evidence of rodent or insect outer openings; written log maintained
19	Garbage Disposal	Outside garbage waste grease and storage area; clean, rodent proof, all containers covered
20	Rodent Proofing	Inside containers or receptacles: adequate number, properly covered and insect/rodent proof
21	Personnel 7-38-012	Certified Food Manager on site when potentially hazardous foods are prepared and served
22	Food Equipment/Utensil 7-38-030	Dish machines provided with accurate thermometers, chemical test kits and suitable gauge cock
23	Equipment Sanitation	Dishes and utensils flushed, scraped, soaked
24	Equipment/Utensil 7-38-030, 005 (A)	Dish washing facilities: properly designed, constructed, maintained, installed, located and operated
25	Toxic Items 7-38-005 (A), 7-38-030	Toxic items properly stored, labeled and used
26	Hand Washing Facilities 7-38-030	Adequate number, convenient, accessible, properly designed and installed
27	Toilet Facilities	Toilet rooms enclosed clean, provided with hand cleanser, sanitary hand drying devices and proper waste receptacles
28	Inspection Report Summary 7-42-010 (B)	Inspection report summary displayed and visible to all customers
29	Compliance	Previous minor violation(s) corrected (7-42-090)

Minor Violations (Items 30–44) — $250 fine if not corrected by next routine inspection

Code	Category	Description
30	Food Labeling	Food in original container, properly labeled; customer advisory posted as needed
31	Utensils	Clean multi-use utensils and single service articles properly stored; no reuse of single service articles
32	Food/Non-Food Surfaces	Food and non-food contact surfaces properly designed, constructed and maintained
33	Equipment Cleanliness	Food and non-food contact equipment utensils clean, free of abrasive detergents
34	Food Maintenance	Floors: constructed per code, cleaned, good repair, covering installed, dust-less cleaning methods used
35	Building Maintenance	Walls, ceilings, attached equipment constructed per code: good repair, surfaces clean and dust-less cleaning methods

36	Lighting	Lighting: required minimum foot-candles of light provided, fixtures shielded
37	Toilet Room Doors	Toilet room doors self closing: dressing rooms with lockers provided; complete separation from living/sleeping quarters
38	Ventilation	Ventilation: rooms and equipment vented as required; plumbing installed and maintained
39	Linen Storage	Linen: clean and soiled properly stored
40	Refrigeration	Refrigeration and metal stem thermometers provided and conspicuous
41	Premises Cleanliness	Premises maintained free of litter, unnecessary articles, cleaning equipment properly stored
42	Ice Handling	Appropriate method of handling of food (ice) hair restraints and clean apparel worn
43	Dispensing Utensils	Food (ice) dispensing utensils, wash cloths properly stored
44	Authorized Personnel	Only authorized personnel in the food-prep area
70	No Smoking	No Smoking Regulations 7-32-010 Through 090

2.5 Chicago Results Classification

The form defines four possible outcomes:

Result	Definition	Derived Score
Pass	No critical or serious violations	90
Pass with Conditions	Critical violations corrected during inspection	80
Fail	Critical or serious violations not corrected during inspection	70
No Entry	Inspector could not access the establishment	0
All other types	Not Ready, Out of Business, Business Not Located, etc.	NULL

3. Dallas (Texas) Inspection Report

The Dallas Retail Food Establishment Inspection Report is issued under the Texas Department of State Health Services (Form EH-06, Revised 09-2015). The form uses a 47-item compliance checklist with a point-based scoring system.

3.1 Report Structure

The Dallas/Texas inspection report consists of three pages. Page 1 contains establishment identification, purpose of inspection, and the full 47-item compliance checklist organized into three tiers (Priority Items, Priority Foundation Items, and Core Items). Page 2 provides corrective actions reference for food safety violations. Page 3 captures temperature observations and detailed observations/corrective actions noted by the inspector.

3.2 Issuing Authority

Texas Department of State Health Services, administered locally by the City of Dallas. The form follows Texas state health code requirements for retail food establishment inspections.

3.3 Scoring System

Dallas uses a point-deduction scoring system starting from a perfect score of 100. Points are deducted based on violation severity:

Violation Tier	Items	Points Deducted	Correction Deadline
Priority Items	1–20	3 points each	Immediate (not to exceed 3 days)
Priority Foundation Items	21–33	2 points each	Within 10 days
Core Items	34–47	1 point each	90 days or next inspection

3.4 Report Fields to Dataset Mapping

Report Field	Dataset Column	Data Type	Notes
Establishment Name	Restaurant_Name	string	Name of the food establishment
Physical Address	Street_Address	string	Full street address
City/County	Not in dataset	—	Hardcoded as DALLAS in Silver
Zip Code	Zip_Code	string	5-digit or ZIP+4 format
Date	Inspection_Date	date	Inspection date
License/Permit #	Not in dataset	—	On form but not captured in dataset
Est. Type	Not in dataset	—	Establishment type (not captured)
Risk Category	Not in dataset	—	On form but not captured in dataset

Purpose of Inspection	Inspection_Type	string	Compliance, Routine, Field Investigation, Visit, Other
TOTAL/SCORE	Inspection_Score	integer	Final numeric score (0–100)
Violation Items (1–47)	Violation_Description_1..25	string	Up to 25 violations captured in wide columns
Violation Points	Violation_Points_1..25	integer	Points per violation (1, 2, or 3)
Violation Detail	Violation_Detail_1..25	string	Specific detail of what was observed
Violation Memo	Violation_Memo_1..25	string	Inspector comments/notes
Contact/Owner Name	Not in dataset	—	On form but not captured
Phone	Not in dataset	—	On form but not captured
Follow-up (Yes/No)	Not in dataset	—	On form but not captured
Temperature Observations	Not in dataset	—	Page 3; not captured in dataset
Inspector Signature	Not in dataset	—	On form but not captured

3.5 Dallas Violation Code Reference (47-Item Checklist)

The Texas DSHS 47-item checklist is organized into three tiers. Violation codes in the dataset correspond to item numbers on this checklist.

Priority Items (3 Points) — Items 1–20

Code	Category	Description
1	Time & Temperature	Proper cooling time and temperature
2	Cold Holding	Proper cold holding temperature (41°F/45°F)
3	Hot Holding	Proper hot holding temperature (135°F)
4	Cooking	Proper cooking time and temperature
5	Reheating	Proper reheating procedure for hot holding (165°F in 2 hours)
6	Time as PHC	Time as a Public Health Control; procedures & records
7	Approved Source	Food and ice from approved source; food in good condition, safe, unadulterated
8	Receiving	Food received at proper temperature
9	Cross-Contamination	Food separated & protected during food prep, storage, display, and tasting
10	Surface Sanitation	Food contact surfaces and returnables; cleaned and sanitized
11	Disposition	Proper disposition of returned, previously served or reconditioned food
12	Employee Health	Management, food employees: knowledge, responsibilities, and reporting
13	Restriction/Exclusion	Proper use of restriction and exclusion; no discharge from eyes, nose, mouth
14	Handwashing	Hands cleaned and properly washed; gloves used properly
15	Bare Hand Contact	No bare hand contact with ready-to-eat foods or approved alternate method
16	Susceptible Populations	Pasteurized foods used; prohibited food not offered
17	Food Additives	Food additives approved and properly stored; washing fruits & vegetables
18	Toxic Substances	Toxic substances properly identified, stored, and used
19	Water/Plumbing	Water from approved source; plumbing installed; proper backflow device
20	Sewage Disposal	Approved sewage/wastewater disposal system

Priority Foundation Items (2 Points) — Items 21–33

Code	Category	Description

21	Person in Charge	Person in charge present, demonstration of knowledge; Certified Food Manager
22	Food Handler	Food handler trained; no unauthorized persons
23	Hot/Cold Water	Hot and cold water available; adequate pressure, safe
24	Records/Labeling	Required records available (shellstock tags); packaged food labeled
25	Approved Procedures	Compliance with Variance, Specialized Process, and HACCP plan
26	Consumer Advisory	Posting of consumer advisories; raw or undercooked foods (disclosure/reminder)
27	Cooling Method	Proper cooling method used; equipment adequate to maintain product temperature
28	Date Marking	Proper date marking and disposition
29	Thermometers	Thermometers provided, accurate, and calibrated
30	Permit	Food establishment permit (current & valid)
31	Handwashing Facilities	Adequate handwashing facilities: accessible and properly supplied
32	Contact Surfaces	Food and non-food contact surfaces cleanable, properly designed, constructed, and used
33	Warewashing	Warewashing facilities; installed, maintained, used; service sink provided

Core Items (1 Point) — Items 34–47

Code	Category	Description
34	Pest Control	No evidence of insect contamination, rodent, or other animals
35	Personal Cleanliness	Personal cleanliness; no eating, drinking or tobacco use
36	Wiping Cloths	Wiping cloths; properly used and stored
37	Environmental	Environmental contamination
38	Thawing	Approved thawing method
39	Utensils	Utensils, equipment, & linens; properly used, stored, dried, handled
40	Single-Service	Single-service & single-use articles; properly stored and used
41	Food ID	Original container labeling (bulk food)
42	Non-Food Surfaces	Non-food contact surfaces clean
43	Ventilation/Lighting	Adequate ventilation and lighting; designated areas used
44	Garbage/Refuse	Garbage and refuse properly disposed; facilities maintained
45	Physical Facilities	Physical facilities installed, maintained, and clean
46	Toilet Facilities	Toilet facilities; properly constructed, supplied, and clean
47	Other	Other violations not covered above

4. Cross-City Report Comparison

The following table compares key structural differences between the Chicago and Dallas inspection report forms and their implications for the data pipeline.

Aspect	Chicago	Dallas
Issuing Authority	City of Chicago Dept. of Public Health	Texas Dept. of State Health Services
Violation Code System	44 items (codes 1–44, 70)	47 items (codes 1–47)
Violation Severity Tiers	Critical ($500), Serious ($250/day), Minor ($250)	Priority (3 pts), Priority Foundation (2 pts), Core (1 pt)
Scoring Method	Categorical (Pass/Fail/etc.)	Numeric (0–100, deduction-based)
Result on Form	Stamp: PASSED / PASSED w/COND / FAILED / CANCELED	Derived from score
Unique Identifier	Inspection # printed on form	No unique ID on form
Risk Category	Printed on form (Risk 1/2/3)	On form but NOT in dataset
Facility Type	Printed on form	Est. Type on form but NOT in dataset
License/Permit	License # on form AND in dataset	License/Permit # on form but NOT in dataset
Inspector Details	Badge # fields on form	Signature fields on form
Temperature Log	Food Temperature Observations section	Temperature Observations page
Data Format	Violations in single text column	Up to 25 violation groups in wide columns

5. Key Observations from Report Analysis

1. Data capture gaps: Both cities' datasets omit significant fields from their inspection forms (inspector details, temperature observations, HACCP information, contact/owner names). This is a systemic limitation of the open data portals, not a data quality issue.

2. Violation code systems are independent: Chicago uses codes 1–44 and 70 based on Chicago municipal code, while Dallas uses codes 1–47 based on Texas DSHS standards. There is no need to create a cross-reference between the two systems.

3. Scoring philosophy differs fundamentally: Chicago uses a categorical pass/fail system where the result depends on whether critical violations were corrected during the inspection, while Dallas uses a numeric deduction system where each violation subtracts points from 100. These differences should be considered when comparing inspection outcomes across cities.

4. Dallas wide-column violation format (25 groups of 4 columns) limits the maximum violations per inspection to 25, whereas the Chicago pipe-delimited format has no practical limit.

5. Chicago's violation text contains rich embedded information (code number, description, and inspector comments in a single string), while Dallas stores these as separate structured columns.

6. Neither dataset captures the inspection form's compliance status markings (IN/OUT/NO/NA/COS/R columns on the Dallas form; checkmarks on the Chicago form). Only the violation descriptions and scores/results are digitized in the open data portals.